Improving search result relevance with emotional modelling and social engagement metrics

Final Assignment CS152

Abstract

This paper proposes an improved user experience for web search through the incorporation of emotional modelling into search result ranking, as a case study for exploring the role of cognitive modelling to create artificial intelligence algorithms that can have higher utility. First, the need for current use cases and potential applications of cognitive modelling in artificial intelligence agents and algorithms is discussed. Then, these arguments are examined in the specific context of Google's web search engine. Existing research about cognitive modelling in AI is discussed, then 2 novel approaches of implementing cognitive modelling are discussed - the first through examining affinity of emotional and personal entities and category of the result, and the second through examining the quantity and quality of social engagement. For each of these approaches, the overview, intuition, high-level system design and limitations are discussed. The emotional and personal affinity approach is then tested on a limited domain through, and the responses received through Pagerank and the emotional and personal affinity approach are tested.

Background - Artificial Intelligence

Artificial Intelligence is the field of creating algorithms that can model or outperform human intelligence. Russell & Norvig categorize Artificial Intelligence into 4 quadrants - thinking rationally, acting rationally, thinking humanly and acting humanly. Generally, thinking rationally simplifies to following logical syllogisms, acting rationally refers to making the choice that yields the highest expected value, even in the absence of complete information, acting humanly refers to passing the Turing test and thinking humanly refers to modelling the human thought process through cognitive and emotional modelling, and deriving a conclusion based on this model.

Currently, large sections of artificial intelligence are dedicated to logical modelling and decision making. Examples of this include machine learning, image classification and recommender systems. These are oriented towards problems with logical outcomes, such as creating autonomous cars and automating decision-making in various fields. However, even within these fields, as more progress is made towards an eventual working solution, there are increasing concerns around ethical concerns. For example, if a self-driving car has to make a choice between killing it's passenger or 2 jaywalkers, which would it choose? Many of these ethical dilemmas are difficult to logically model because human decision-making is highly emotional but tends to be justified after-the-fact with logical reasoning that does not hold up to further scrutiny. This phenomenon of how humans think has been described as a "emotional dog with a logical tail" by scholars. Therefore, research on emotional modelling could prove highly useful to creating machines that are able to navigate more ambiguous decisions.

While we have discussed the applications of emotional modelling in situations with logical outcomes, cognitive modelling has significant and immediate implications for several consumer-facing products and problems where the problems do not have clear logical outcomes. Current applications of emotional modelling in artificial intelligence are in creating chatbots and in synthesizing market research. Other situations with ambiguous outcomes where emotional modelling could improve outcomes are content recommendation, human resource performance optimization and any sort of integration between artificially intelligent agents and human agents in natural language environments - ranging from therapy to mentorship to teaching. Advancements in such integrations would greatly increase the adoption of artificial intelligence agents in highly personal and personalized environments.

Background - Search Engine

Currently, Google's search engine primarily utilizes a pagerank algorithm to determine the credibility of a website. A web page's Pagerank can be summarized as a product of the quantity and quality of incoming links into that website. Pagerank is supplemented by over 200 other factors such as social media engagement and domain diversity. However, according to Dan Russell, a senior search scientist at Google, Google does not currently conduct any emotional modelling of either the query or the results. This does not hold any discernible effects when modelling queries with logical responses. In this case, Pagerank serves as an efficient dictionary, serving the logical answer to the question. However, it leads to heavy confirmation bias or just an imperfect search result when applied to queries that are more personal, emotional or ambiguous where the user may not know exactly what he or she is looking for. This ineffectiveness is apparent in 2 use cases familiar to the author - queries about health (mental and physical) and queries about advice.

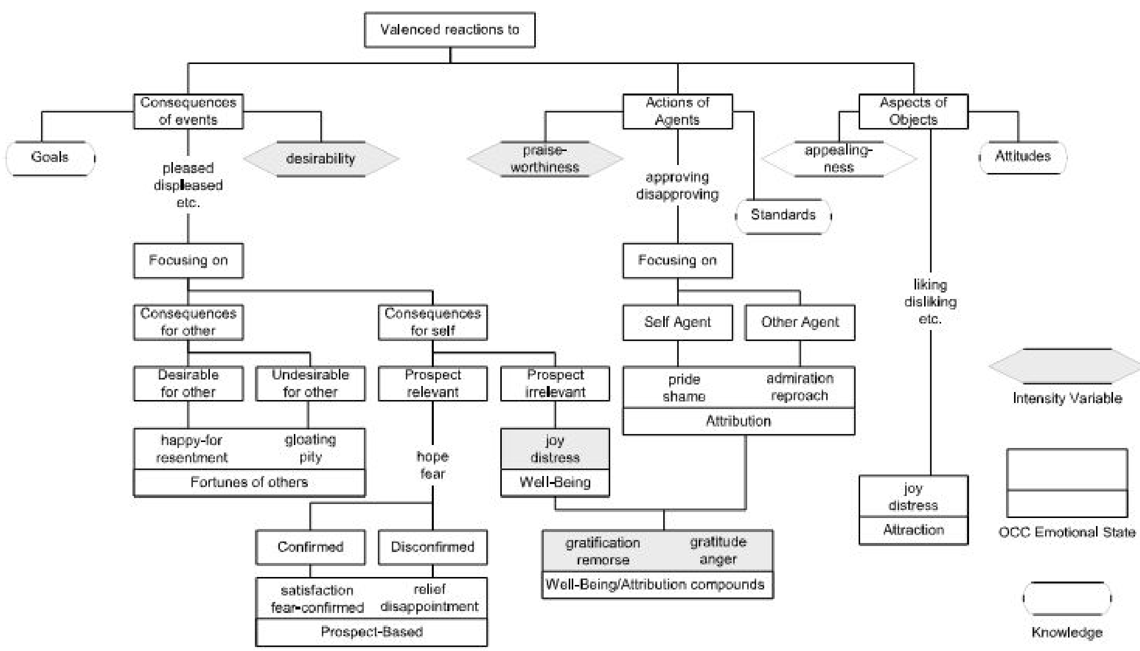
In health-related queries, Google's search engine returns results that are tangentially related but suggest conditions that are often significantly worse than reality. For example, a search on "Why do I feel tired" returns "nemia, diabetes, hypothyroidism, hepatitis C, sleep apnea, obstructive sleep apnea, chronic fatigue syndrome, urinary tract infection, food sensitivities, heart disease, depression, anxiety disorder, and nasal congestion" as possible reasons. This is a result of the logic-centered pagerank approach to delivering a result with the highest quantity*quality of incoming links. Similarly, queries related to advice, such as "Should I go to grad school", return highly general results that do not take into consideration the user's major, country, city or income level, and thus provide highly unspecific results. The user will need to search several variations of the search in order to piece together the information that would help him or her.

Cognitive Modelling can be illustrated with both the above examples A cognitive modelling approach to the "Why do I feel tired" query would establish the user's state of mind through triangulating his/her demographic, environment and other data points the user has explicitly allowed the algorithm to use, and then derive the cognitive process behind the question to answer it. To provide a concrete example in terms of a single variable, in the case of this query about being tired, if it were to be asked at 3pm, results related to sleep apnea should have higher value, but if it were to be asked at 7am, results related to sleep quality should be prioritized.

A cognitive modelling to the “Should I go to grad school” query would similarly establish the user’s context - what is his or her major, income and geographic area - and come up with an answer for *why* the user is asking that query - for example which academic area is it in, is it based on professional interest or academic interest? This would then allow results that address the thoughts that led to the query - for example a possible result that could be returned in this model would be about an alternative to grad school. Therefore, emotional modelling in this instance provides the user much more valuable set of information rather than a general set of results that directly addresses the question.

Cognitive Modelling

There are several frameworks for Cognitive modelling. The “OCC” Model is a model developed by Ortony, Clore and Collins models emotions and established itself as the standard model for emotion synthesis. It specifies 22 emotion categories based on reactions to situations that are either goal relevant events, acts of an accountable agent or as attractive or unattractive objects.



However, this model lacks the specificity to incorporate the multiple nuanced emotions individuals may feel. For example, there are only 3 categories to classify emotions towards objects, and this greatly generalizes the wide spectrum of emotions humans have towards objects. Moreover, this framework merely maps emotions but fails to deconstruct or attempt to model the environment that led to that emotion, which is important to effective cognitive modelling.

There are other popular forms of cognitive modelling, such as utilizing neural networks, that form a bottom-up approach by modelling individual neurons. However, these models have not been trained by centuries of evolution or the human lived experience and hence do not display the same complex intuition or emotional responses.

This paper suggests 2 new models of modelling emotional affinity and engagement. First, a measure of categorization and entity recognition is used to provide results that are similarly categorized and contain similar entities. Second, a measure of social emotional engagement is proposed.

The overall model will first qualify the query with background information that will be gleaned from the user's profile. Results will be similarly categorized. Within each category, emotional entities will be recognized and ranked. Results will be returned based on these emotional entities through a search mechanism. The content of the page itself is examined for these entities. Then, an extension of this algorithm that analyzes social emotional engagement is also put forward and analyzed.

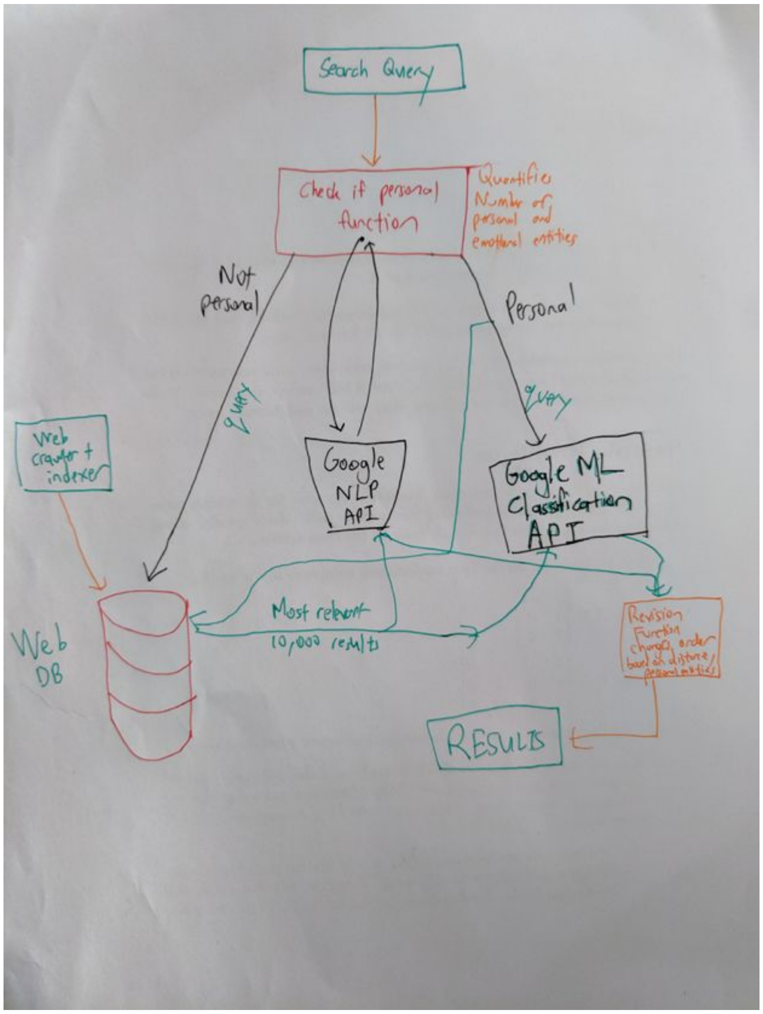
Examining affinity of emotional and personal entities

The first step is qualification of the search query. The algorithm only caters to queries that require emotional modelling, as queries that have clear logical answers are presently served well by the existing pagerank algorithm.

However, current research is highly concentrated on categorizing text as either positive or negative, but does not break this down into more refined emotional states. Even this analysis has been limited to cases such as one-line headlines or other datasets of small size and limited scope (Gaint, Syal & Padgalwar). There is no dataset of emotion words that are labelled according to emotion categories. Gaind, Syal & Padgalwar algorithmically created a dataset of words in 6 emotional categories - Happiness, Sadness, Anger, Surprise, Fear and Disgust, and trained an algorithm that can detect these entities. This is only a small subset of human emotion, however, and therefore more research needs to be done to achieve an algorithm that can recognize emotional entities in text. Another step is to quantify personal entities in the search queries. A simple way this can be done is to identify words that are related to the sense of self, such as "I", "my", "me". The quantity of personal entities as well as the distance between the query and result emotional categories in an emotional category tree will be factors that can supplement Google pagerank.

Then, the results will be put through a similar algorithm to identify affinity of emotional and personal entities. Affinity of emotional entities can be recognized through a similar process of training a classifier that can identify words or phrases that represent emotion-based entities. Quantifying personal entities is slightly more comprehensive as terms that are related to not only the sense of self but also to persons in general could be deduced to have content that is personal. For example, a career advice blog or a story about a person struggling with a career change both are stories that are relevant as they both contain personal entities.

Concretely, the overall system design is as follows.



Social Engagement

Another key difference between logical and emotional content is the differing effects of having content that effectively addresses the intended purpose of the website visitor. Pagerank works well for Logical content because writing effective content creates credibility and increases the number of incoming links. However, this may not hold for emotional content because the sharing and expression of emotional content tends to take place in private social networks that are not indexable by the web.

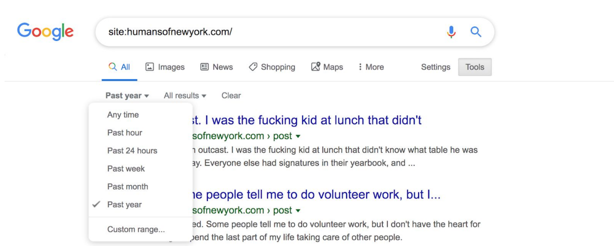
Therefore, audience engagement in the form of quantity of sharing on social media as well as the quantity and quality of audience interaction, in the form of comments. It is assumed that a post that effectively engages audience on an emotional level will solicit comments, and therefore the quantity of comments will be taken as a factor. Moreover, beyond quantity, the quality of the response in terms of length, popularity and affinity of emotional entities to the main post can also be examined, in order to quantify the emotional engagement of commenters. The design and implementation of this mechanism is outside the scope of this paper.

Constraints

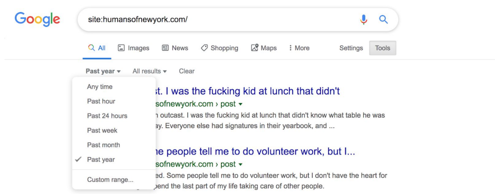
Constraints of the first method - examining affinity of emotional and personal entities - are that there does not exist an existing database or classification algorithm that can comprehensively identify and categorize emotion-based entities or person-based entities. There are some frameworks that can be used, but these frameworks are heavily limited as they only allow gauging individual words for emotions, not entire texts. General sentiment analysis algorithms provide an overall positive or negative sentiment without isolating specific words that convey emotion, which can be used as entities. This implementation therefore utilizes Google's Natural Language Processing API to get the entities present. Google's NLP API returns broad categories but allows the training of a custom model that better fits our intended outcome, but that is outside the scope of this course. Another constraint introduced by the lack of available data on emotional modelling is that a central premise of the overall system design - content recommendation based on distance from the general emotional category of the query from the general emotional category of the result. This implementation utilizes Google's Content Classification algorithm to get a general category of content, and the search results implement the metrics outlined in this paper. Although outside the scope of this course, a future paper for a machine learning course would implement trained models with learning and test data for both of these problems, in order to create a more specific implementation.

Analysis of results

Given the constraints outlined above, a limited search algorithm was implemented. The sample query was 'Why do I feel tired all the time'. Google's search was limited to the Humans of New York website from the past year. This was done by inserting the query parameter site:humansofnewyork.com and limiting the time range.



However, it was discovered that the pagerank mechanism returned no results.



A sample of 5 queries and the number of results returned are listed below.

Query	**Results**
Why do I feel tired	0
Exam Stress	0
I feel happy	0
I felt sad	0
Why do I feel stressed	0

The author attempted to replicate the pagerank algorithm but unfortunately was not able to do so. However, each of the functions proposed was implemented and tested as shown in the accompanying python notebook. It is not possible to implement these functions without access to a search algorithm, which is difficult to create from scratch, but the independent tests do show that each individual function works, and therefore the overall system design proposed can be implemented and tested with more work in machine learning and access to an open source search engine. The author attempted to limit his sample but this led to large tradeoffs in terms of quality of results returned.

<u>Conclusion</u>

Artificial intelligence has so far been dominated by modelling logic. This paper takes a deep dive into an unexplored area of Artificial Intelligence - emotional modelling - and examines the background, use cases and potential in the future. A case study in web search was explored, with several new mechanisms suggested for qualifying search queries as emotional or logical, quantifying search results by social emotional engagement and presence of personal and emotional entities. These mechanisms were designed and implemented with the exception of the metric for social emotional engagement. Current search was explored and analyzed and the constraints for integrating a solution were discovered in terms of a lack of emotion classification frameworks and lack of open source search frameworks that allowed limiting search queries. Further research could expand on the ideas explored towards the concrete goal of creating better search and eventually more emotionally intelligent artificial intelligence.

LOs and HCs

#aiconcepts - A novel approach was taken to examining emotional modelling in Artificial Intelligence. Concepts from AI, Psychology and ethics are integrated to present a persuasive argument for further research on creating AI algorithms that think humanly, as described by Russell & Norvig. Prior research was examined, the implications for the current and future state of AI were discussed and a new model for incorporating emotional modelling in search engines was proposed and then implemented in a limited manner within the constraints of lack of data, prior research and flexible open source web search frameworks.

#search - The limitations of the existing graph search algorithm utilized by Google, Pagerank, were examined and several new functions were proposed and created in order to optimize for emotional intelligence. How these algorithms fit into the current informed search algorithm used by Google by adding a dimensionality of personal and emotional entity recognition and modelling is also discussed.

#thesis - A well structured thesis was written with a clear problems statement that is addressed and referred back to throughout the paper.

#emergentproperties - A nuanced discussion of the implications of Artificial Intelligence with and without Emotional Intelligence is undertaken and used to justify further research into creating emotional models.

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